

- (e) Ethanoic acid is a member of the homologous series of carboxylic acids.

State the general formula of this homologous series.

..... [1]

- (f) Draw the structure of the carboxylic acid containing three carbon atoms. Show all of the atoms and all of the bonds.

[2]

- (g) When carboxylic acids react with alcohols, esters are produced.

The formula of ester **X** is $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_3$.

- (i) Name ester **X**.

..... [1]

- (ii) Give the name of the carboxylic acid and the alcohol that react together to produce ester **X**.

carboxylic acid

alcohol

[2]

- (h) Ester **Y** has the following composition by mass:

C, 48.65%; H, 8.11%; O, 43.24%.

Calculate the empirical formula of ester **Y**.

empirical formula = [3]

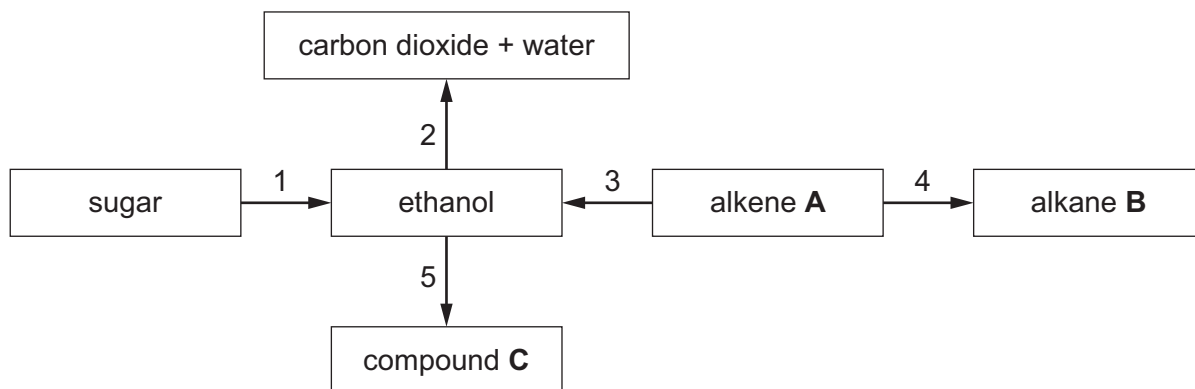
- (i) Ester **Z** has the empirical formula $\text{C}_2\text{H}_4\text{O}$ and a relative molecular mass of 88.

Determine the molecular formula of ester **Z**.

molecular formula = [1]

[Total: 19]

5 The reaction scheme shows five organic reactions, numbered 1 to 5.



(a) Name reaction 1.

..... [1]

(b) Name reaction 2 and write the chemical equation for this reaction.

name

equation [3]

(c) Reaction 3 forms ethanol from alkene **A**.

(i) Identify alkene **A**.

..... [1]

(ii) State the type of reaction that occurs during reaction 3.

..... [1]

(iii) State the reagents and conditions needed for reaction 3.

.....

..... [2]

(d) Alkene **A** is converted into alkane **B** in reaction 4.

(i) State the reagent and conditions for reaction 4.

.....

..... [3]

(ii) State the general formula of alkanes.

..... [1]

(e) Ethanol is oxidised in reaction 5 by heating it with dilute sulfuric acid and one other reagent.

(i) Identify the other reagent in reaction 5.

..... [1]

(ii) Name the homologous series compound **C** belongs to.

..... [1]

(iii) Draw the structure of compound **C**.

Show all of the atoms and all of the bonds.

[1]

[Total: 15]

6 This question is about polymers.

(a) Polymer **X** is a condensation polymer.

Part of the structure of polymer **X** is shown.



(i) How many molecules of water are produced when this part of polymer **X** is formed from its monomers?

..... [1]

(ii) Complete the structures of the **two** monomers used to make polymer **X**.

Show all of the atoms and all of the bonds in the functional groups.



and



[2]

(iii) What type of condensation polymer is **X**?

..... [1]

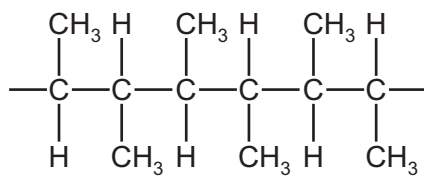
(b) Part of polymer **Y** has the structure shown.



State the number of different types of monomer needed to make polymer **Y**.

..... [1]

(c) Part of polymer **Z** has the structure shown.



(i) Draw and name the structure of the monomer which forms polymer **Z**.

Show all of the atoms and all of the bonds.

name

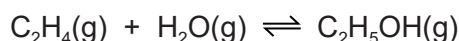
[3]

(ii) Name the chemical process used to make the monomer that forms polymer **Z**.

..... [1]

[Total: 9]

- 4 Ethanol is made industrially by the reaction of ethene with steam. The reaction occurs at a temperature of 300 °C and a pressure of 60 atmospheres.



A catalyst is used in this reaction.

The forward reaction is exothermic.

- (a) State what is meant by the term *catalyst*.

.....
 [2]

- (b) Complete the table using only the words *increases*, *decreases* or *no change*.

	effect on the rate of the forward reaction	effect on the equilibrium yield of C ₂ H ₅ OH(g)
increasing the temperature		
decreasing the pressure		

[4]

- (c) Ethanol is a member of the alcohol homologous series.

Members of the same homologous series have the same general formula.

- (i) State the general formula of alcohols.

..... [1]

- (ii) State two **general** characteristics, other than the same general formula, of all homologous series.

1

2

[2]

- (iii) One alcohol containing three carbon atoms is propan-1-ol.

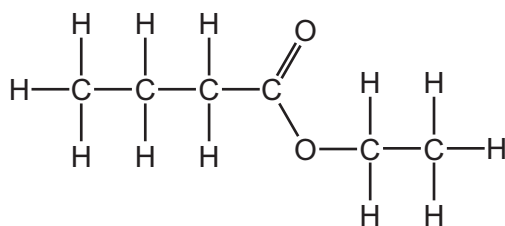
Draw the structure of one **other** alcohol containing three carbon atoms. Show all of the atoms and all of the bonds.

Name the alcohol you have drawn.

name [2]

- (d) When alcohols react with carboxylic acids, esters are produced.

- (i) The structure of ester **X** is shown.



Name ester **X**.

..... [1]

- (ii) Give the name of the alcohol and the carboxylic acid that react together to produce ester **X**.

alcohol

carboxylic acid [2]

- (e) Ester **Y** has the following composition by mass:

C, 58.82%; H, 9.80%; O, 31.37%.

Calculate the empirical formula of ester **Y**.

empirical formula = [3]

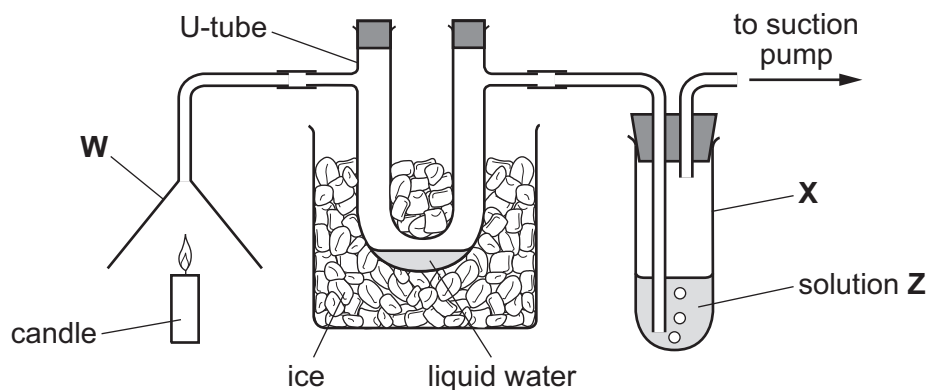
- (f) Ester **Z** has the empirical formula $\text{C}_3\text{H}_6\text{O}$ and a relative molecular mass of 116.

Calculate the molecular formula of ester **Z**.

molecular formula = [1]

[Total: 18]

- 1 The apparatus in the diagram was used to show that when a candle is burned both water and carbon dioxide are formed. The gases produced when the candle burns are passed through the apparatus using a suction pump.



- (a) Name the items of apparatus labelled **W** and **X**.

W

X

[2]

- (b) Suggest why ice is placed around the U-tube.

.....

..... [1]

- (c) Describe how to test the liquid collected in the U-tube to show it is water.

.....

..... [1]

- (d) Solution **Z** is used to show that carbon dioxide is produced.

Identify solution **Z**.

..... [1]

- (e) Both water and carbon dioxide were made.

Identify **one** element that must be in the compound that makes up the candle.

..... [1]

- (f) Describe how the apparatus could be changed to see if sulfur dioxide is made.
Give the observations if sulfur dioxide is made.

change

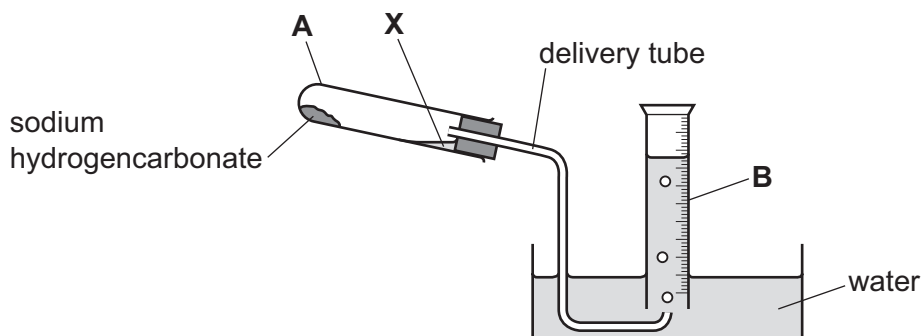
observation

[2]

[Total: 8]

- 1 Sodium hydrogencarbonate decomposes when heated. The products are solid sodium carbonate, water and carbon dioxide.

A student decomposed a sample of sodium hydrogencarbonate using the apparatus shown.



- (a) Name the items of apparatus labelled **A** and **B**.

A

B

[2]

- (b) When the sodium hydrogencarbonate was heated, a colourless liquid collected at the point marked **X**.

Suggest the identity of the colourless liquid.

..... [1]

- (c) **On the diagram** draw one arrow to show where the apparatus should be heated during the experiment. [1]

- (d) State an observation that would indicate the sodium hydrogencarbonate had stopped reacting.

.....

..... [1]

- (e) Explain why it is important to remove the delivery tube from the water as soon as heating is stopped.

.....

.....

..... [2]

[Total: 7]

5 The table shows the names or structures of organic compounds **P** to **U**.

P	Q	R
$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array} $	propanoic acid	but-1-ene
S	T	U
propan-1-ol	methyl butanoate	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \diagdown \quad \quad \\ \text{C}=\text{C}-\text{C}-\text{H} \\ \diagup \quad \quad \\ \text{H} \quad \quad \text{H} \end{array} $

(a) Give the letters of the organic compounds, **P** to **U**, that are unsaturated hydrocarbons.

..... [2]

(b) Describe the test for an unsaturated hydrocarbon.

test

observations

[2]

(c) But-1-ene is an unbranched molecule.

(i) Name the unbranched isomer of but-1-ene.

..... [1]

(ii) Draw the structure of a branched isomer of but-1-ene. Show all of the atoms and all of the bonds.

[1]

(d) Dodecane is an alkane with 12 carbon atoms. Dodecane can be cracked.

(i) Write the formula of dodecane.

..... [1]

(ii) Give the letters of all the organic compounds, **P** to **U**, that can be formed when dodecane is cracked.

..... [2]

- (e) Name the reagent and suggest the conditions needed to convert organic compound **U** into organic compound **S**.

reagent

conditions

[3]

- (f) Organic compound **S** can be converted to organic compound **Q** by reaction with an acidified reagent.

- (i) Name the type of chemical change that happens to organic compound **S**.

..... [1]

- (ii) Name the acidified reagent added to organic compound **S**.

..... [1]

- (g) Organic compound **T** is made by reacting two compounds together.

- (i) Name the homologous series that organic compound **T** belongs to.

..... [1]

- (ii) Name the **two** compounds which react together to make organic compound **T**.

Draw the structures of each compound you have named. Show all of the atoms and all of the bonds.

name

structure

name

structure

[4]

- (iii) Deduce the molecular formula of organic compound **T**.

..... [1]

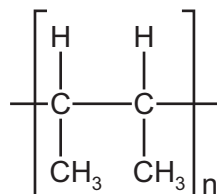
[Total: 20]

6 Polymers are large molecules built up from small molecules.

(a) State the name given to the small molecules from which polymers are made.

..... [1]

(b) The formula of a polymer is shown.



(i) Draw the structure of the small molecule from which this polymer is made. Show all of the atoms and all of the bonds.

[2]

(ii) State the type of polymerisation used to make this polymer.

..... [1]

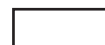
(c) Three amino acids are shown. They combine to form part of a natural polymer.



(i) Name the type of natural polymer formed when amino acids combine.

..... [1]

(ii) Complete the diagram to show part of the structure of the natural polymer that forms when these three amino acids combine. Show all of the bonds in the linkages.



[3]

(iii) Name the type of chemical reaction that takes place when this natural polymer is converted back to amino acids.

..... [1]

[Total: 9]

7 Many organic compounds contain carbon, hydrogen and oxygen only.

(a) An organic compound **V** has the following composition by mass.

C, 48.65%; H, 8.11%; O, 43.24%

Calculate the empirical formula of compound **V**.

empirical formula = [3]

(b) Compound **W** has the empirical formula CH_4O and a relative molecular mass of 32.

Calculate the molecular formula of compound **W**.

molecular formula = [1]

(c) Compounds **X** and **Y** have the same general formula.

X and **Y** are both carboxylic acids.

Compound **X** has the molecular formula $\text{C}_2\text{H}_4\text{O}_2$.

Compound **Y** has the molecular formula $\text{C}_4\text{H}_8\text{O}_2$.

(i) Deduce the general formula of compounds **X** and **Y**.

..... [1]

- (ii) Draw the structure of compound Y. Show all of the atoms and all of the bonds.

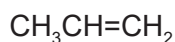
Name compound Y.

name [3]

- (iii) Give the name used to describe a 'family' of similar compounds with the same general formula, similar chemical properties and the same functional group.

..... [1]

- (d) Propene is an unsaturated hydrocarbon. The formula of propene is shown.



- (i) State the colour change observed when propene is added to aqueous bromine.

from to [1]

- (ii) Propene can be produced by cracking long chain alkanes.

Pentadecane, $\text{C}_{15}\text{H}_{32}$, is cracked to produce an alkane and propene in a 1 : 2 molar ratio.

Complete the chemical equation for this reaction.

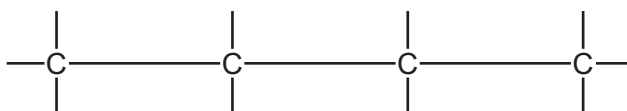


- (iii) Propene can be converted into poly(propene).

Name the type of polymerisation that occurs when propene is converted into poly(propene).

..... [1]

- (iv) Complete the diagram to show a section of poly(propene).



[2]

[Total: 15]

6 Molecules **A** and **B** can form condensation polymers.



(a) Each molecule has two identical functional groups.

(i) Name the functional group in **B**.

..... [1]

(ii) Draw the part of the structure of the synthetic polymer that would form when two molecules of **A** and two molecules of **B** combine. Show all of the bonds in the linkages.

[3]

(iii) Name the other product formed when molecules of **A** and **B** undergo polymerisation.

..... [1]

(b) Molecule **A** is a simple sugar unit which can be made by hydrolysis of complex carbohydrates.

(i) Draw part of the complex carbohydrate that could be hydrolysed to make molecules of **A**.

Include **one** linkage and show all of the bonds in the linkage.

[1]

(ii) State **two** sets of conditions which could be used to hydrolyse the complex carbohydrate to form **A**.

1

2

[2]

(iii) Name the technique used to identify the individual sugar units made by the hydrolysis of a complex carbohydrate.

..... [1]

(c) Ethanol can be made from the simple sugar glucose, $C_6H_{12}O_6$.

(i) State the name of this process.

..... [1]

(ii) Complete the chemical equation for this reaction.



[Total: 12]

7 Many organic compounds contain carbon, hydrogen and oxygen only.

(a) An organic compound **R** has the following composition by mass.

C, 69.77%; H, 11.63%; O, 18.60%

Calculate the empirical formula of compound **R**.

empirical formula = [2]

(b) Compound **S** has the empirical formula CH_2O and a relative molecular mass of 60.

Calculate the molecular formula of compound **S**.

molecular formula = [2]

(c) Compounds **T** and **V** have the same molecular formula, $\text{C}_3\text{H}_6\text{O}_2$.

- Compound **T** is an ester.
- Compound **V** contains a $-\text{COOH}$ functional group.

(i) State the name given to compounds with the same molecular formula but different structures.

..... [1]

(ii) Name the homologous series that **V** is a member of.

..... [1]

(iii) Draw a structure of compound **T**. Show all of the atoms and all of the bonds.

Name compound **T**.

name [3]

(iv) Draw the structure of compound **V**. Show all of the atoms and all of the bonds.

Name compound **V**.

name [2]

(d) Ethanol can be produced from long chain alkanes such as decane, $C_{10}H_{22}$, in a two-step process.



For each of the two steps:

- name the type of chemical reaction that occurs
- write a chemical equation.

step 1: decane to ethene

type of reaction

chemical equation

step 2: ethene to ethanol

type of reaction

chemical equation

[4]

[Total: 15]

- 5 Alkenes and carboxylic acids are both families of similar compounds with similar chemical properties. Alkenes and carboxylic acids have different reactions.

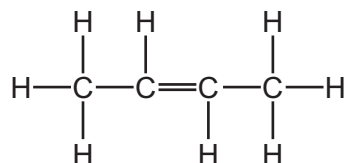
(a) State the term used for a 'family' of similar compounds.

..... [1]

(b) State the general formula of alkenes.

..... [1]

(c) The structure of but-2-ene is shown.



(i) But-2-ene reacts with aqueous bromine in an addition reaction.

Describe the colour change seen when but-2-ene is added to aqueous bromine.

from to [1]

(ii) State what is meant by the term *addition reaction*.

..... [1]

(iii) Write the chemical equation for the reaction between but-2-ene and bromine.

..... [2]

(iv) But-2-ene forms a polymer.

Suggest the name of the polymer formed from but-2-ene.

..... [1]

(v) Name and draw a structural isomer of but-2-ene.

Show all of the atoms and all of the bonds.

name

structure

[2]

(d) Butanoic acid, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$, is a carboxylic acid.

(i) Deduce the empirical formula of butanoic acid.

..... [1]

(ii) Complete the chemical equation for the reaction of butanoic acid and sodium carbonate, Na_2CO_3 .

$2\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow \dots + \dots + \dots$ [2]

(iii) Butanoic acid reacts with methanol to form an organic compound and water.

- Name the organic compound formed.

..... [1]

- Draw the structure of the organic compound formed.

Show all of the atoms and all of the bonds.

[2]

[Total: 15]

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6 Ethanol, $\text{C}_2\text{H}_5\text{OH}$, belongs to the homologous series called alcohols.

(a) Write the general formula of alcohols.

..... [1]

(b) Explain why ethanol **cannot** be described as a hydrocarbon.

..... [1]

(c) Ethanol can be manufactured from different substances by reaction with steam or by fermentation.

(i) Give the formula of the substance which reacts with steam to form ethanol.

..... [1]

(ii) Name a substance which will undergo fermentation to form ethanol.

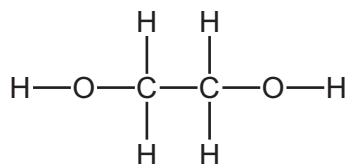
..... [1]

(d) Ethanol is a fuel.

Write the chemical equation for the complete combustion of ethanol.

..... [2]

- (e) Ethane-1,2-diol has two alcohol functional groups.



One molecule of ethane-1,2-diol will react with two molecules of ethanoic acid to form molecule **X**.

X has two ester functional groups and a molecular formula of $\text{C}_6\text{H}_{10}\text{O}_4$.

- (i) State the empirical formula of **X**.

..... [1]

- (ii) Draw the structure of **X**.

Show all of the atoms and all of the bonds.

[2]

- (iii) Name the **other** substance formed in this reaction.

..... [1]

- (f) Each alcohol functional group in ethane-1,2-diol reacts with acidified potassium manganate(VII) to form a different organic compound, **Y**.

- (i) Name the functional groups formed in **Y**.

..... [1]

- (ii) Draw the structure of **Y**.

Show all of the atoms and all of the bonds.

[1]

[Total: 12]

- 7 (a) Ethanol is a member of the homologous series of alcohols.

Give **two** characteristics of members of a homologous series.

1

2 [2]

- (b) Ethanol can be manufactured from ethene.

Ethene can be made from long chain hydrocarbons such as decane, $C_{10}H_{22}$.

Ethene is then converted into ethanol.

- (i) Name the process used to obtain ethene from long chain hydrocarbons such as decane, $C_{10}H_{22}$.

..... [1]

- (ii) Complete the chemical equation to show the formation of ethene from decane, $C_{10}H_{22}$.



- (iii) Write the chemical equation for the conversion of ethene into ethanol.

..... [1]

- (iv) Name the type of reaction occurring when ethene is converted into ethanol.

..... [1]

- (v) Give **one** condition for the reaction in which ethene is converted into ethanol.

..... [1]

- (c) Ethanol can also be produced by fermentation of carbohydrates such as glucose.

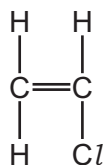
Give **two** advantages of manufacturing ethanol by fermentation compared to manufacturing ethanol from ethene.

1

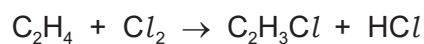
2 [2]

- (d) (i) Under certain conditions ethene can react with chlorine to produce chloroethene.

The structure of chloroethene is shown.



The equation for the chemical reaction is shown.



State the type of chemical reaction between ethene and chlorine that this equation shows.

..... [1]

- (ii) Chloroethene monomers can be converted into a polymer called poly(chloroethene).

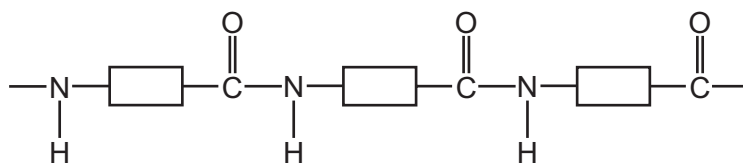
State the type of polymerisation that produces poly(chloroethene) from chloroethene.

..... [1]

- (iii) Draw a section of the poly(chloroethene) molecule made from **two** monomer molecules.

[2]

(e) The structure of part of a polymer is shown.



This polymer is made from one type of monomer only.

Complete the diagram to show the structure of the monomer used to produce this polymer. Show all of the atoms and all of the bonds in the functional groups.

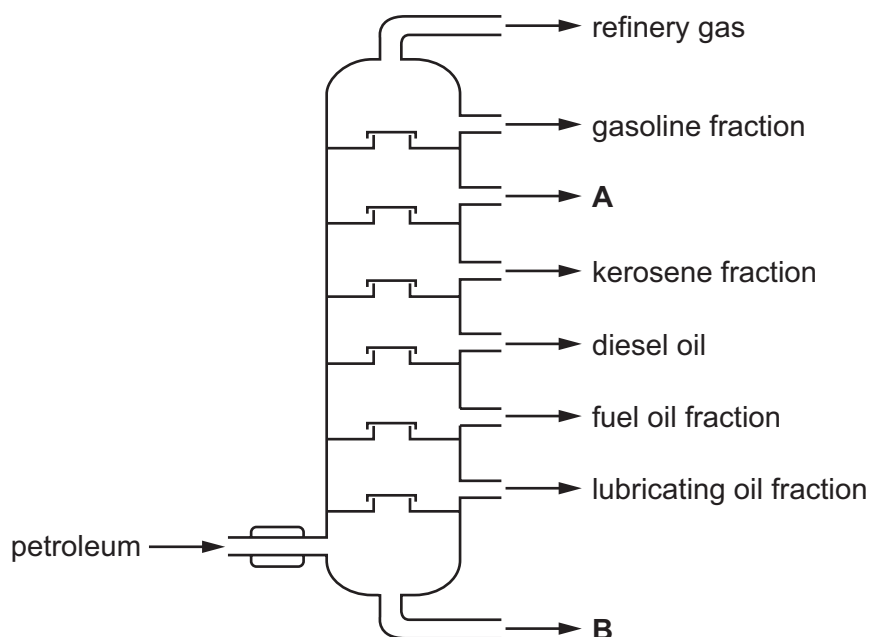


[2]

[Total: 16]

1 Petroleum is a useful natural resource.

The diagram shows how petroleum can be separated into useful substances.



(a) What is the name of the separation process shown in the diagram?

..... [2]

(b) Name the fraction leaving at:

A

B [2]

(c) Refinery gas is a mixture of hydrocarbons.

One refinery gas is butane, C_4H_{10} .

(i) Suggest the names of **two** other refinery gases.

..... and [2]

(ii) Write the chemical equation for the complete combustion of butane.

..... [2]

(iii) Name the toxic gas produced by the incomplete combustion of butane.

..... [1]

(d) Gasoline and kerosene are both fuels. They have different properties.

(i) Describe the differences in the properties given.

viscosity of the fuel

.....

flammability of the fuel

.....

[2]

(ii) What difference in the molecules of gasoline and kerosene causes these differences in properties?

..... [1]

(e) Hydrogen fuel cells can be used to power vehicles.

Write the word equation for the overall reaction that takes place in a hydrogen fuel cell.

..... [1]

[Total: 13]

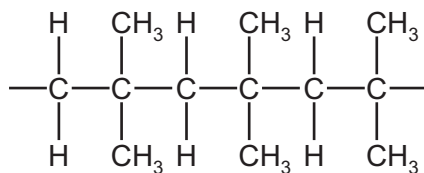
5 There are two types of polymers.

(a) Addition polymers are made from many identical small units.

(i) What is the term used to describe these small units?

..... [1]

(ii) A section of an addition polymer is shown.



Draw the structure of the small unit used to make this addition polymer.

Show all of the atoms and all of the bonds.

[2]

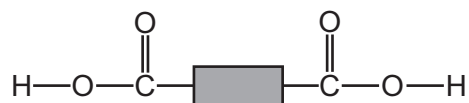
(b) Polyamides are condensation polymers.

What does the term *condensation* mean when used to describe this type of polymer?

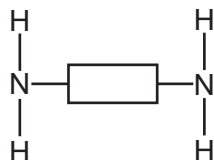
..... [1]

- (c) A polyamide can be made from two different molecules.

A simplified structure of octanedioic acid is shown.



A simplified structure of 1,6-diaminohexane is shown.



- (i) Complete the diagram to show a section of polyamide manufactured from octanedioic acid and 1,6-diaminohexane. Include all of the atoms and all of the bonds in the linkages.



[3]

- (ii) State the name of a synthetic polyamide.

..... [1]

[Total: 8]

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5 Ethanol is manufactured by two different processes.

(a) For each process, name the organic reactant and state the type of reaction.

organic reactant type of reaction

organic reactant type of reaction

[4]

(b) Alcohols can be oxidised to form carboxylic acids.

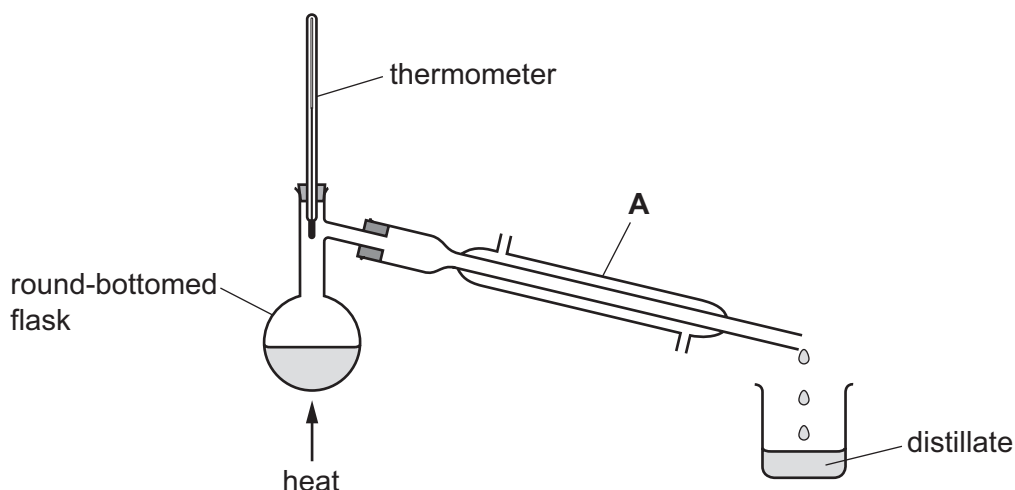
Name a suitable oxidising agent for this reaction.

..... [1]

(c) Alcohols can be partially oxidised to form aldehydes.

Aldehydes are a homologous series of organic compounds.

Partial oxidation is achieved by reacting an alcohol with the oxidising agent in distillation apparatus as shown.



(i) Name apparatus **A**.

..... [1]

(ii) On the diagram, use **one** arrow to show where water enters apparatus **A**.

[1]

(d) The table shows some information about aldehydes.

(i) Complete the table.

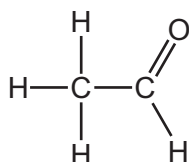
name		ethanal	propanal	butanal
molecular formula	CH ₂ O	C ₂ H ₄ O	C ₃ H ₆ O	

[2]

(ii) Deduce the general formula of aldehydes.

..... [1]

(e) The structural formula of ethanal is shown.

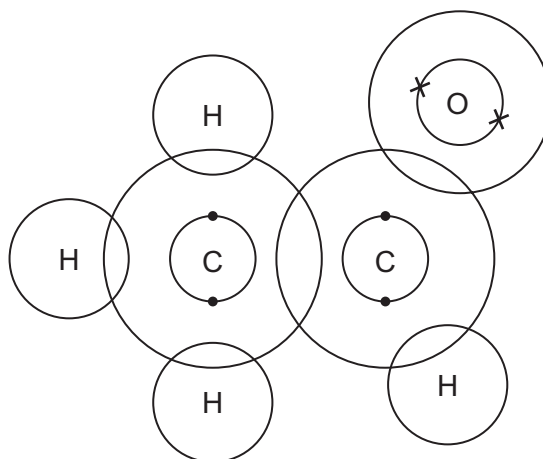


The C=O group in aldehydes is at the end of the carbon chain.
This is a reactive part of the molecule.

(i) What is the name given to the reactive part of any organic molecule?

..... [1]

(ii) Complete the dot-and-cross diagram to show the electron arrangement of a molecule of ethanal. Inner shells have been drawn.



[3]

(f) Propanone belongs to a homologous series called ketones. Ketones have the same C=O group as aldehydes but the C=O group is not at the end of the carbon chain. Propanone has the same molecular formula as propanal, C_3H_6O .

(i) What term is used to describe molecules with different structures but with the same molecular formula?

..... [1]

(ii) Suggest the structure of propanone, C_3H_6O . Show all of the atoms and all of the bonds.

[2]

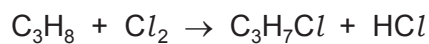
[Total: 17]

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- 6 (a) Propane reacts with chlorine in a photochemical reaction as shown.



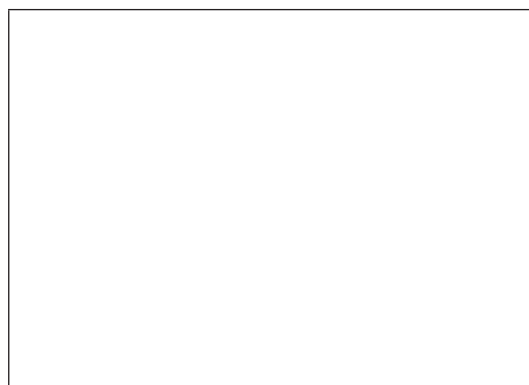
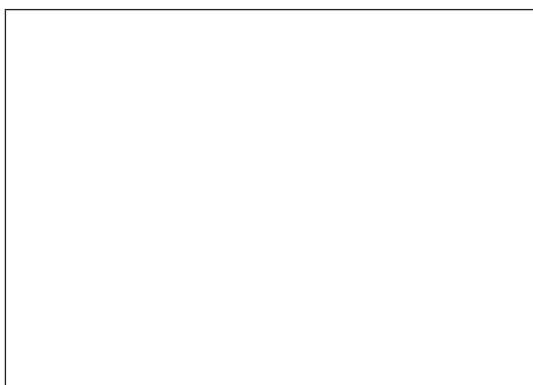
- (i) What type of reaction is this?

..... [1]

- (ii) What condition is needed for this photochemical reaction to occur?

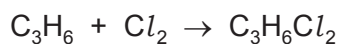
..... [1]

- (iii) Draw **two** structural isomers of compounds with the formula $\text{C}_3\text{H}_7\text{Cl}$.
Show all of the atoms and all of the bonds.



[2]

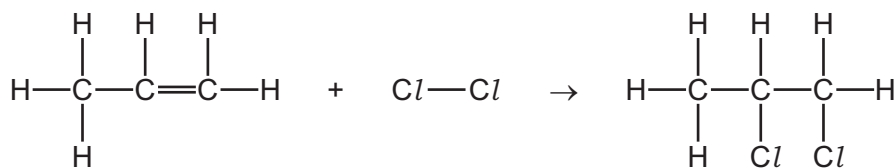
- (b) Propene reacts with chlorine in an addition reaction as shown.



- (i) State why this is an addition reaction.

..... [1]

(ii) The structures of the reactants and products of this reaction are shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
C–C	347
C=C	612
C–H	413
C–Cl	339
Cl–Cl	242

Calculate the energy change for the reaction between propene and chlorine using the following steps.

- Calculate the energy needed to break the bonds.

..... kJ

- Calculate the energy released when bonds are formed.

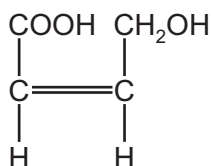
..... kJ

- Calculate the energy change for the reaction between propene and chlorine.

..... kJ/mol
[3]

(c) There are three functional groups in compound **A**.

compound **A**



(i) Name the homologous series of compounds that contains the following structures.

C=C

–OH

–COOH

[3]

(ii) What would you observe when compound **A** is added to:

aqueous bromine

aqueous sodium carbonate?

[2]

(d) Compound **A** can be used as a single monomer to produce two different polymers.

(i) Draw **one** repeat unit of the addition polymer formed from compound **A**.

[2]

(ii) What type of condensation polymer is formed from compound **A**?

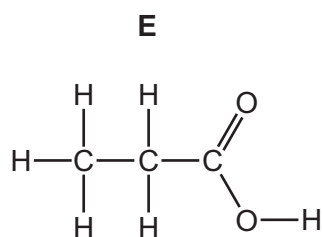
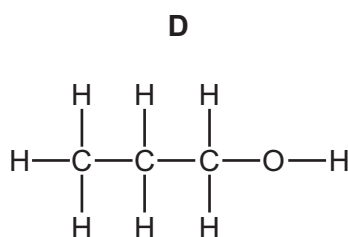
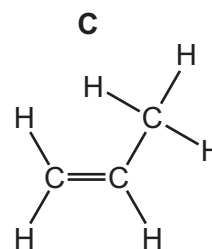
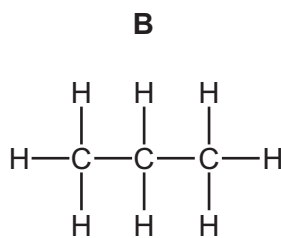
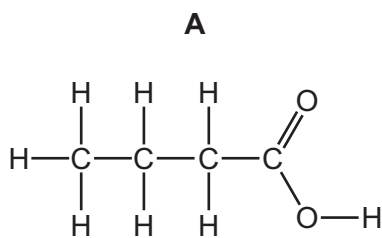
..... [1]

[Total: 16]

- 1 (a) The structures of five organic compounds, **A**, **B**, **C**, **D** and **E**, are shown.

Answer the questions that follow.

Each letter may be used once, more than once or not at all.



- (i) Give the letter of the compound that is propan-1-ol.

..... [1]

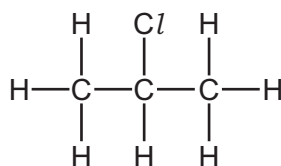
- (ii) Give the letter of the compound that has the empirical formula CH_2 .

..... [1]

- (iii) Give the letter of **one** compound that reacts with bromine in an addition reaction.

..... [1]

- (iv) Give the letter of **one** compound that reacts with chlorine to form the compound shown.



..... [1]

- (v) Give the letters of **two** compounds that can react with each other to form an ester.

..... and [1]

- (vi) Give the letter of the compound that is in the same homologous series as hex-1-ene.

..... [1]

- (vii) Give the letter of **one** compound that is an acid.

..... [1]

(viii) Draw a structural isomer of compound **D**.

Show all of the atoms and all of the bonds.

[1]

(b) Some acids are described as weak acids.

State the meaning of the term *weak acid*.

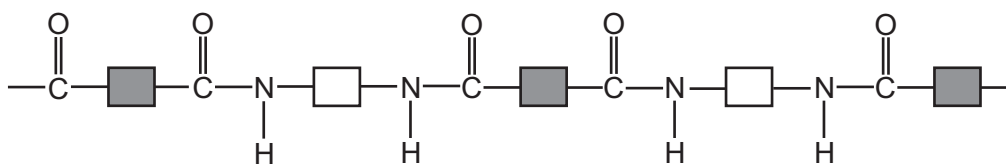
weak

acid

[2]

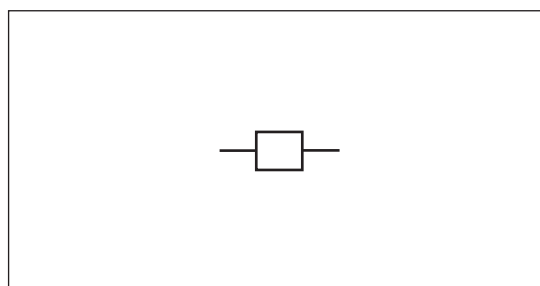
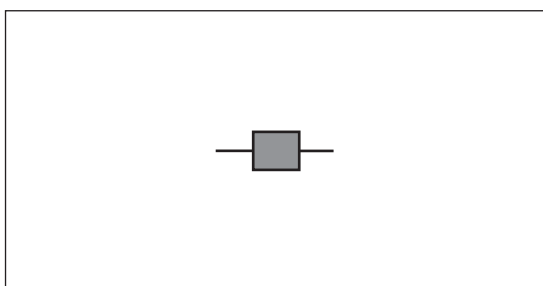
[Total: 10]

- 8 (a) Part of the synthetic polymer, nylon, is shown in the diagram.



- (i) Circle **one** amide linkage **on the diagram**. [1]

- (ii) Complete the structures of the **two** monomers that react to form nylon.



[2]

- (iii) Name the other product formed when nylon is produced.

..... [1]

- (b) Items made from nylon are often disposed of by burying them in the ground. This is called landfill.

Why is the disposal of nylon using landfill a problem?

.....
 [1]

- (c) Give the name of a natural polymer.

..... [1]

[Total: 6]

6 (a) Ethane, propane and butane are members of the same homologous series.

(i) Name this homologous series.

..... [1]

(ii) State **two** ways members of the same homologous series are similar.

1

2 [2]

(b) One mole of ethane, C_2H_6 , contains 6.02×10^{23} molecules.

Calculate how many molecules are in 15 g of ethane.

number of ethane molecules = [1]

(c) Propane reacts with chlorine.

(i) Write the formula of the product which does not contain carbon.

..... [1]

(ii) Draw the structure of an organic product formed. Show all of the atoms and all of the bonds.

[1]

(iii) State the name of this type of reaction.

..... [1]

- (d) (i) Aqueous bromine was added to a sample of ethene.

Give the colour change seen.

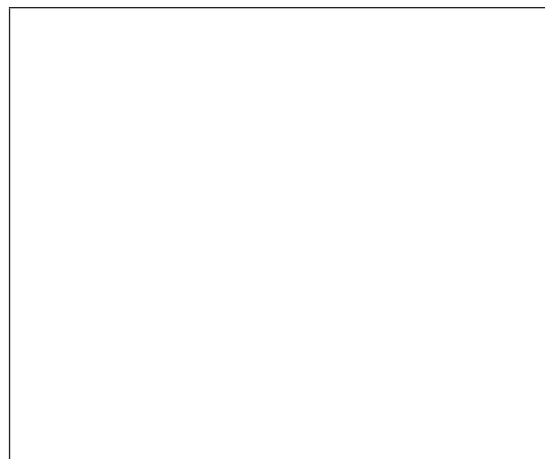
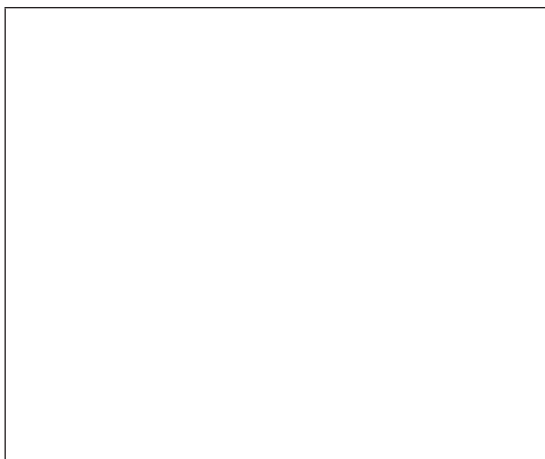
from to [2]

- (ii) Explain, in terms of bonding, why there is no colour change when aqueous bromine is added to ethane.

.....
..... [1]

- (e) There are two structural isomers with the formula C_4H_{10} .

- (i) Draw the structures of both of these isomers, showing all of the atoms and all of the bonds.



[2]

- (ii) Butane is formed when longer chain hydrocarbons are cracked.

Complete the chemical equation to show the other product when butane is formed by cracking.



(f) A compound contains 85.7% carbon and 14.3% hydrogen by mass.

(i) Calculate the empirical formula of this compound.

Show your working.

..... [2]

(ii) The molecular mass of the compound is 112.

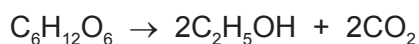
Calculate the molecular formula of this compound.

..... [1]

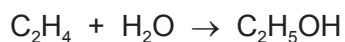
[Total: 16]

- 7 (a) Ethanol can be manufactured by two different methods.

Method 1: fermentation of a sugar, $C_6H_{12}O_6$



Method 2: reaction of ethene with steam



- (i) Give **one** advantage of using fermentation compared with Method 2.

.....
 [1]

- (ii) Give **one** disadvantage of using fermentation compared with Method 2.

.....
 [1]

- (b) Ethanol reacts with acidified potassium manganate(VII) to form water and a product that turns litmus red.

- (i) State the name of the product that turns the litmus red.

..... [1]

- (ii) State the type of reaction that ethanol undergoes when it reacts with acidified potassium manganate(VII).

..... [1]

- (c) Ethanol reacts with methanoic acid to form an ester.

- (i) Name the ester formed in this reaction.

..... [1]

- (ii) Draw the structure of the ester formed.
 Show all of the atoms and all of the bonds.

[1]

(d) The table shows the melting points of ethanol and sodium chloride.

substance	melting point/°C
ethanol	−114
sodium chloride	801

The difference in melting points is due to differences in attractive forces between particles in these substances.

Name the type of attractive force in each substance, which is responsible for the difference in melting points.

ethanol

sodium chloride

[2]

[Total: 8]

4 Alkenes and alkanes are homologous series of compounds containing carbon and hydrogen atoms.

(a) State the name of the type of compound made from carbon and hydrogen atoms only.

..... [1]

(b) Alkenes take part in addition reactions.

(i) Describe what is meant by the term *addition reaction*.

..... [1]

(ii) Draw the structure of the product made in the addition reaction between propene and bromine. Show all of the atoms and all of the bonds.

[2]

(iii) Describe the colour change seen when propene is added to aqueous bromine.

from to [2]

(iv) Draw the structures of molecules of **two** different alkenes which both undergo an addition reaction with steam to form butan-2-ol. Show all of the atoms and all of the bonds.

[2]

(c) Propane undergoes a substitution reaction with chlorine.

Write the chemical equation for the reaction between one molecule of propane and one molecule of chlorine.

..... [2]

[Total: 10]

5 This question is about alcohols, carboxylic acids and esters.

(a) Ethanol will react with hot aqueous potassium manganate(VII) to form ethanoic acid.

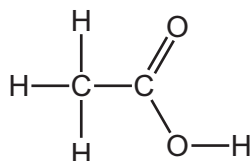
(i) State the other condition needed for this reaction to take place.

..... [1]

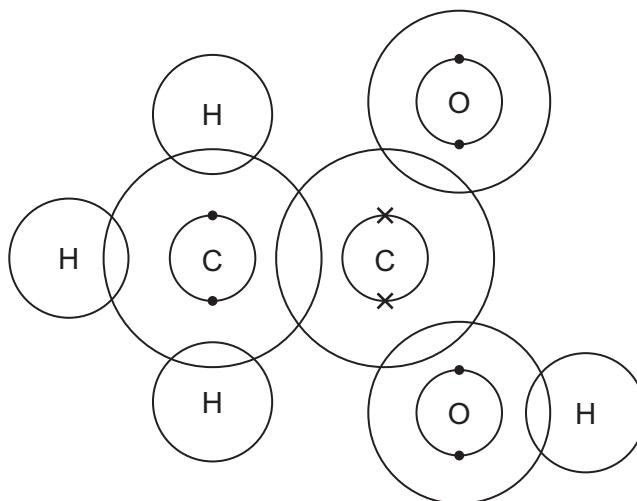
(ii) State the type of chemical change that happens to the ethanol during this reaction.

..... [1]

(iii) The structure of ethanoic acid is shown.



Complete the dot-and-cross diagram to show the electron arrangement in a molecule of ethanoic acid.



[3]

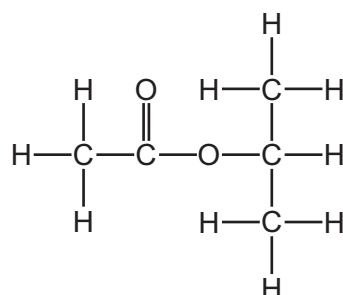
- (b) Ethanoic acid is a weak acid and hydrochloric acid is a strong acid.

Complete the table to show the similarities and differences in the properties of samples of these two acids of equal concentration.

	dilute ethanoic acid	dilute hydrochloric acid
extent of dissociation		
colour after adding universal indicator solution		
observation when magnesium ribbon is added		

[6]

- (c) Ethanoic acid will react with an alcohol to form the ester shown.



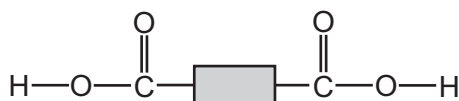
- (i) Name the **other** product formed when ethanoic acid reacts with an alcohol to make this ester.
 [1]
- (ii) Give **one** condition needed when ethanoic acid reacts with the alcohol to make this ester.
 [1]
- (iii) Draw the structure of the alcohol which was added to ethanoic acid to make this ester. Show all of the atoms and all of the bonds.

[2]

- (d) Polyesters can be manufactured from carboxylic acids and alcohols.

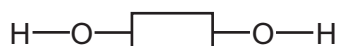
Hexanedioic acid has the structure: $\text{HOOC}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{COOH}$.

This structure can be simplified as shown.



Ethanediol has the structure: $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$.

This structure can be simplified as shown.



The functional groups are found at the end of each molecule.

- (i) State what is meant by the term *functional group*.

..... [1]

- (ii) Determine the empirical formula of hexanedioic acid.

..... [1]

- (iii) Calculate the percentage by mass of oxygen present in ethanediol.

Give your answer to the nearest whole number.

..... % [2]

- (iv) Complete the diagram to show a section of polyester manufactured from hexanedioic acid and ethanediol. Include all of the atoms and all of the bonds in the linkages.



[2]

- (v) State the name of a polyester.

..... [1]

[Total: 22]

6 (a) A carboxylic acid and an ester are structural isomers.

(i) State the meaning of the term *structural isomers*.

..... [2]

(ii) Draw the structures of the carboxylic acid and the ester which both contain two carbon atoms.

Show all of the atoms and all of the bonds.

Name the carboxylic acid and the ester.

carboxylic acid

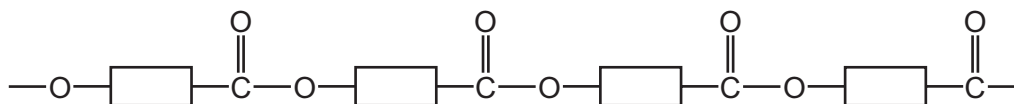
name

ester

name

[4]

(b) Part of a polyester chain is shown. This polyester is made from one monomer.

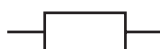


(i) **On the diagram** draw a ring around one unit of the polymer that is repeated. [1]

(ii) Name the type of polymerisation that produces polyesters.

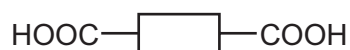
..... [1]

(iii) Complete the diagram to show the structure of the monomer used to produce this polyester. Show all of the atoms and all of the bonds in the functional groups.



[2]

(c) A polyamide is made from the two monomers shown.



Complete the diagram to show a section of the polyamide made from the two monomers. Show all of the atoms and all of the bonds in the linkages.



[2]

(d) Naturally occurring polyamides are constituents of food.

(i) State the name given to naturally occurring polyamides.

..... [1]

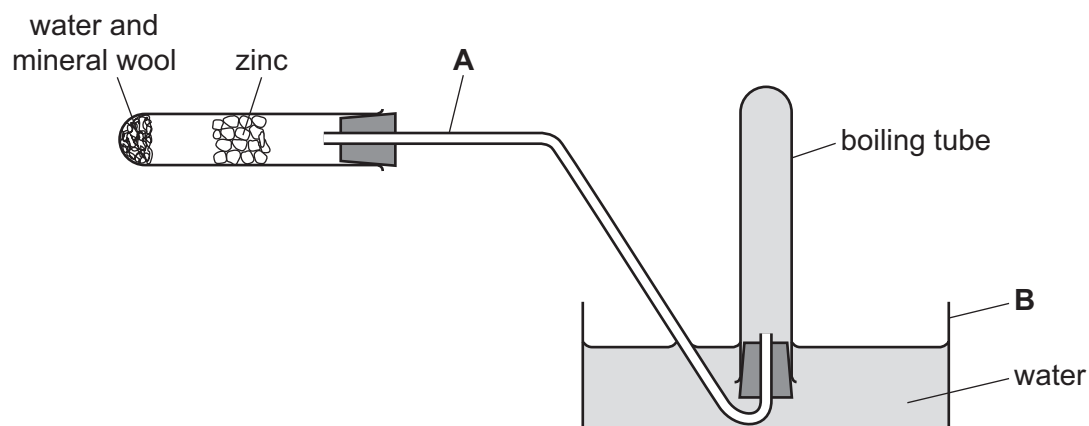
(ii) Name the monomers which form naturally occurring polyamides.

..... [1]

[Total: 14]

- 1 Hot zinc reacts with steam to make zinc oxide and hydrogen gas.

A student wanted to use the apparatus shown to react zinc with steam and to collect the hydrogen.



- (a) Name the items of apparatus labelled **A** and **B**.

A

B

[2]

- (b) State the purpose of the mineral wool.

.....

..... [1]

- (c) The apparatus shown is dangerous to use because of an error in the way it has been set up.

Identify this error.

Explain why this error makes it dangerous to use the apparatus.

error

.....

explanation

.....

[2]

- (f) Tetrafluoroethene is an unsaturated compound with the formula C_2F_4 .
Tetrafluoroethene is the monomer used to make the polymer poly(tetrafluoroethene).

(i) What is meant by the term *unsaturated*?

.....
..... [1]

(ii) Describe a test to show that tetrafluoroethene is unsaturated.

test.....

observations [2]

(iii) Draw the structure of a molecule of tetrafluoroethene. Show all of the atoms and all of the bonds.

[1]

(iv) Tetrafluoroethene can be polymerised to form poly(tetrafluoroethene).

Draw **one** repeat unit of poly(tetrafluoroethene). Show all of the atoms and all of the bonds.

[2]

(v) Deduce the empirical formula of:

tetrafluoroethene

poly(tetrafluoroethene).

[2]

[Total: 20]

(d) Add **two** arrows to the diagram to show the two places where the apparatus should be heated once the error in (c) has been corrected. [1]

(e) Describe the test for hydrogen gas.

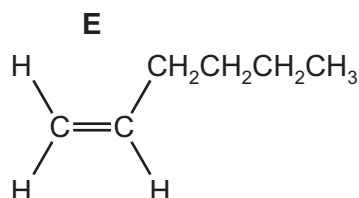
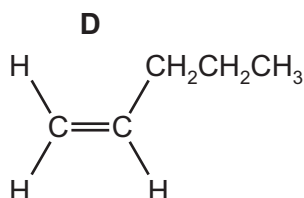
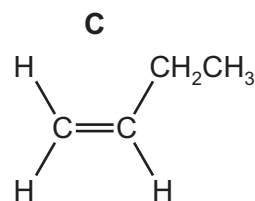
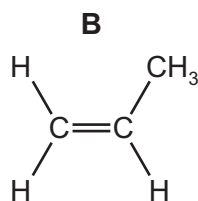
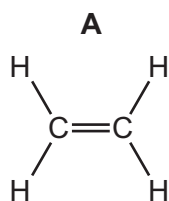
test

result

[2]

[Total: 8]

5 The structures of five alkenes, **A**, **B**, **C**, **D** and **E**, are shown.



(a) What is the general formula of alkenes?

..... [1]

(b) What is the molecular formula of alkene **D**?

..... [1]

(c) Predict which alkene, **A**, **B**, **C**, **D** or **E**, has the highest boiling point.
Explain your answer.

alkene

explanation

..... [2]

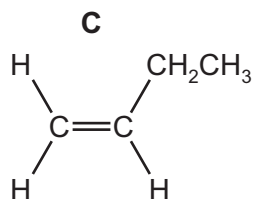
(d) Which alkene, **A**, **B**, **C**, **D** or **E**, diffuses most quickly?
Explain your answer.

alkene

explanation

..... [2]

- (e) A student added aqueous bromine to alkene **C**.



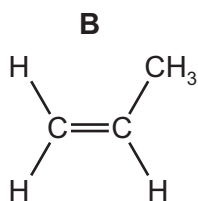
Describe the colour change seen and draw the structure of the product. Show all of the atoms and all of the bonds.

colour change from to

structure

[2]

- (f) Two different alcohols can be produced from alkene **B** by an addition reaction.



- (i) Draw the structures of the **two** alcohols. Show all of the atoms and all of the bonds.

[2]

- (ii) State the reagent and conditions needed to produce an alcohol from alkene **B**.

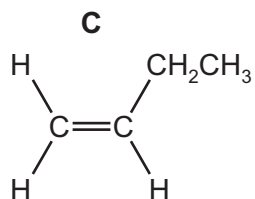
reagent

conditions

.....

[3]

(g) Alkene **C** can be converted into a polymer.



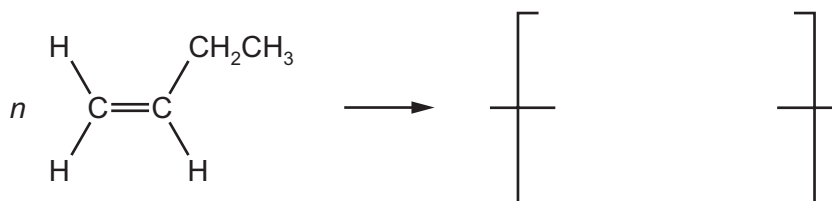
(i) What type of polymerisation occurs?

..... [1]

(ii) Suggest the name of the polymer formed.

..... [1]

(iii) Complete the chemical equation to show this polymerisation.



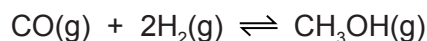
[3]

(iv) State the empirical formula of the polymer formed.

..... [1]

[Total: 19]

- 4 Methanol is made industrially by reacting carbon monoxide with hydrogen. The gases react at a temperature of 250 °C and a pressure of 75 atmospheres.



The forward reaction is exothermic.

- (a) Suggest a source of hydrogen for this industrial process.

..... [1]

- (b) Complete the table using only the words *increases*, *decreases* or *no change*.

	effect on the rate of the reverse reaction	effect on the equilibrium yield of CH ₃ OH(g)
adding a catalyst		no change
increasing the temperature	increases	
decreasing the pressure		

[4]

- (c) Methanol is a member of the homologous series of alcohols.

- (i) State **two** general characteristics of a homologous series.

1

2

[2]

- (ii) Draw the structures of **two** different alcohols, each containing **three** carbon atoms. Show all of the atoms and all of the bonds.

Name these **two** alcohols.

name

name

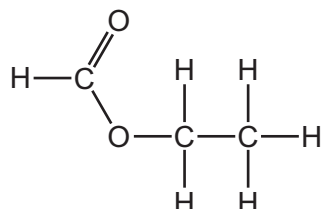
[4]

- (iii) What term is used to describe compounds with the same molecular formula but different structural formulae?

..... [1]

- (d) Alcohols react with carboxylic acids to produce esters.

- (i) The structure of ester **X** is shown.



Name ester **X**.

..... [1]

- (ii) Give the name of the carboxylic acid and the alcohol that react together to produce ester **X**.

carboxylic acid

alcohol

[2]

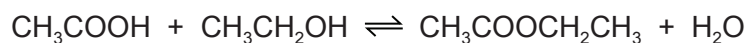
- (iii) Ester **Y** is different from ester **X** but also has the formula $C_3H_6O_2$.

Draw the structure of ester **Y**. Show all of the atoms and all of the bonds.

..... [2]

[Total: 17]

- 5 Carboxylic acids react with alcohols to form esters. The reaction is reversible. The equation for the reaction between ethanoic acid and ethanol is shown.



- (a) (i) What is the name of the ester formed in this reaction?

..... [1]

- (ii) Draw the structure of the ester formed. Show all of the atoms and all of the bonds.

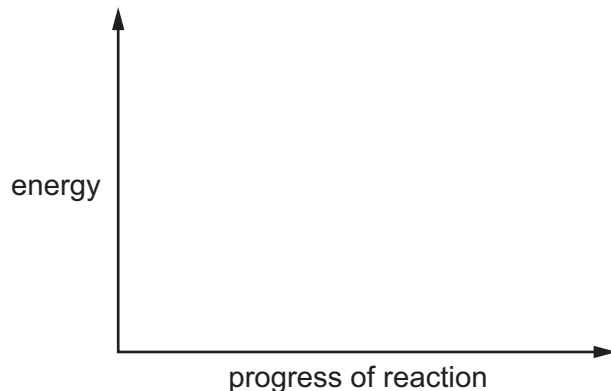
[1]

- (b) The reaction between ethanoic acid and ethanol is exothermic.

Draw an energy level diagram for this reaction.

On your diagram label:

- the reactants and products
- the energy change of the reaction, ΔH .



[3]

- (c) Concentrated sulfuric acid is a catalyst for this reaction.

What is meant by the term *catalyst*?

.....
 [2]

- (d) The rate of reaction can be increased by increasing the temperature.

Explain why increasing the temperature increases the rate of reaction.

.....

.....

.....

.....

.....

..... [4]

- (e) The reaction between ethanoic acid and ethanol reaches equilibrium.

- (i) The reaction between ethanoic acid and ethanol is exothermic.

State and explain the effect, if any, of increasing the temperature on the amount of ester at equilibrium.

.....

.....

..... [2]

- (ii) State and explain the effect, if any, of removing water from the mixture on the amount of ester at equilibrium.

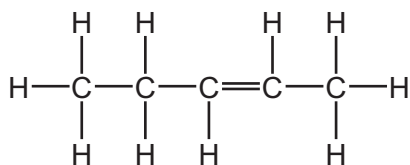
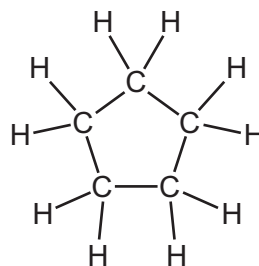
.....

.....

..... [2]

[Total: 15]

- 6 (a) Two hydrocarbons have the structures shown.

hydrocarbon **A**hydrocarbon **B**

- (i) Why are these **two** compounds *hydrocarbons*?

.....
 [2]

- (ii) Hydrocarbon **B** reacts in the same way as a typical alkane.

Describe a chemical test to tell the difference between hydrocarbon **A** and hydrocarbon **B**.

State the name of the reagent you would use and the result you would obtain with hydrocarbon **A** and hydrocarbon **B**.

reagent

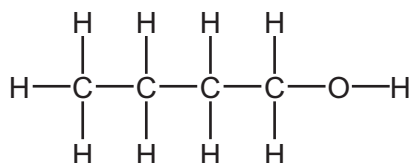
result with hydrocarbon **A**

result with hydrocarbon **B**

[3]

- (b) Alkenes react with steam to form alcohols.

Compound **C** is an alcohol.

compound **C**

Draw the structure of the alkene which could be reacted with steam to make compound **C**.
 Show all of the atoms and all of the bonds.

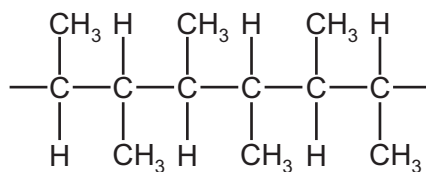
[1]

(c) Alkenes can form polymers.

(i) What type of polymerisation occurs when alkenes form polymers?

..... [1]

(ii) Part of the structure of a polymer is shown.

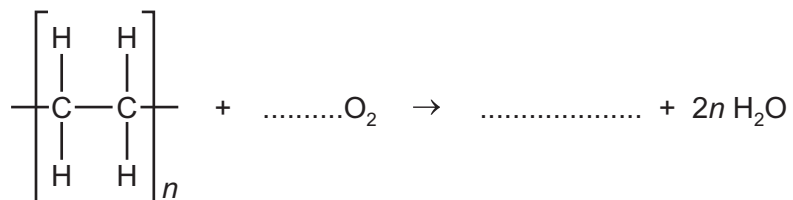


Draw the structure of the alkene from which this polymer can be made. Show all of the atoms and all of the bonds.

[1]

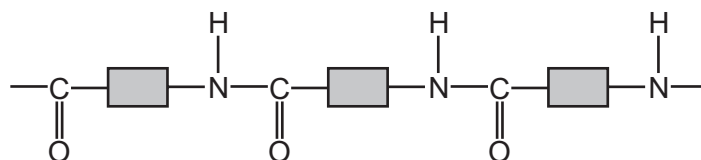
(iii) Polymers can undergo incomplete combustion to form carbon monoxide.

Complete the chemical equation for the incomplete combustion of poly(ethene). The only carbon-containing product is carbon monoxide.



[2]

(d) Part of the structure of a polyamide is shown.



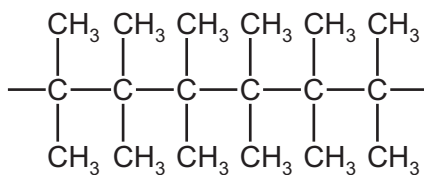
This polyamide is formed from identical monomers. Complete the diagram to show the structure of **one** monomer. Show all of the atoms and all of the bonds.



[2]

[Total: 12]

- 5 (a) Part of the structure of synthetic polymer **A** is shown.



- (i) What type of synthetic polymer is **A**?

..... [1]

- (ii) Deduce the empirical formula of polymer **A**.

..... [1]

- (iii) Draw the structure of the monomer from which polymer **A** is made.

[2]

- (b) The formula C_4H_{10} represents two different structural isomers.

- (i) What is meant by the term *structural isomers*?

.....

 [2]

- (ii) Draw the structures of **two** structural isomers with the formula C_4H_{10} .
 Show all of the atoms and all of the bonds.

[2]

- (iii) All structural isomers of C_4H_{10} are flammable.

Write a chemical equation for the **incomplete** combustion of C_4H_{10} .

..... [2]

[Total: 10]

7 This question is about ethanol.

- (a) Ethanol that is suitable for use as a fuel can be manufactured from sugars such as glucose, $C_6H_{12}O_6$, by a two-step process.

Describe how this can be done. In your answer, include:

- an equation for the reaction in which ethanol is formed
- the essential conditions for the reaction in which ethanol is formed
- the name of the process used to obtain ethanol that is pure enough to use as a fuel from the reaction mixture.

.....

.....

.....

.....

.....

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.....

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.....

.....

..... [5]

(c) Methanol reacts with propanoic acid to form an ester with a molecular formula $C_4H_8O_2$.

(i) Name the ester formed when methanol reacts with propanoic acid.

..... [1]

(ii) Name **one** other substance formed when methanol reacts with propanoic acid.

..... [1]

(iii) Draw the structure of an ester which is a structural isomer of the ester named in (c)(i).
Show all of the atoms and all of the bonds.

[3]

(iv) State the conditions needed to form an ester from a carboxylic acid and an alcohol.

..... [1]

[Total: 12]

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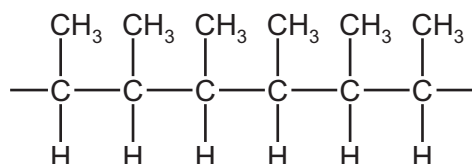
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7 Addition polymerisation and condensation polymerisation are two types of polymerisation.

- (a) Which functional group is present in all the monomers which are used to make addition polymers?

..... [1]

- (b) Part of an addition polymer is shown.



- (i) How many monomer units are needed to make the part of the addition polymer shown?

..... [1]

- (ii) Draw the structure of the monomer that is used to make this addition polymer. Show all of the atoms and all of the bonds.

Name the monomer.

name [2]

- (iii) State the empirical formula of:

the monomer

the polymer. [2]

- (c)** Complex carbohydrates are natural condensation polymers. They can be broken down into colourless monomers which can then be separated and identified.

X is a complex carbohydrate.

Starting with a sample of **X**, describe how to produce, separate, detect and identify the monomers which make it up.

Your answer should include:

- the name of the process used to break down **X** into its monomers
- two** types of substance that can be used to break down **X**
- the name of the process used to **separate** the monomers
- the method used to **detect** the monomers after they have been separated
- the method used to **identify** the monomers after they have been separated and detected.

[6]

- (d)** Synthetic polyamides are condensation polymers.

- (i) Name a synthetic polyamide.

..... [1]

- (ii) Synthetic polyamides can be made by reacting carboxylic acids with amines.

Name the other substance that is produced in this reaction.

..... [1]

[Total: 14]